Maël Conejos Crusson [redacted]

# School of Informatics & Cybersecurity - Assessment Submission Coversheet

## Student Information

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|---|---|
| Student Number: | [redacted] |
| Student Name: | Maël Conejos Crusson |

## Module Information

| | |
|---|---|
| Module Title: | Database Fundamentals |
| Module Code: | COMP H1029 |

## Assessment Information

| | |
|---|---|
| Assessment Title: | DBF_Project |

## Declaration

☑ I confirm that I have read and understood the University's Academic Integrity Policy.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/academic-quality-assurance-and-enhancement/academic-integrity/

☑ I have completed the TU Dublin AI Driving License theory test.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/lta/lta-resources/artificial-intelligence-driving-license/

☑ I declare that this assessment submission is my own work.

☑ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

| | |
|---|---|
| **Student signature:** | [redacted] |
| **Date:** | 20/11/2025 |

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# Database Fundamentals Project

## Task One – Database Design

### Relevant Entities (UNF)

#### Strong

- **department** (deptId (PK), deptname, location, deptPhoneNumber)
- **treatment** (treatId (PK), treatname, treatStatus)
- **staff** (staffId (PK), staffFirstName, staffLastName, jobTitle, hireDate, daysOfWork, drStatus)
- **patient** (patId (PK), patFirstName, patLastName, patPhoneNumber, emailAddress, homeAddress)
- **appointment** (appId (PK), appDate, appTime, patId (FK), treatId (FK)) - strong

#### Weak

- **medical_history** (patientId (PK, FK))

### Explanation

The 'department' entity needs to be included because it divides the hospital into several sections that each are linked to their respective treatments. The 'department' entity holds useful information like location, which is useful for staff and patients. It also shows the department name as well as a department phone number to reach one another easily. And it naturally has a unique department ID for each department to uniquely identify each one of them.

The 'treatment' entity is also needed because it specifies the reason as to why a patient would be in the hospital. It also establishes the patient's relation to the staff through appointments. It contains a treatment ID, a treatment name, and a treatment status. I added the status since it specifies which treatment is in which state: a treatment can be upcoming, undergoing, completed, or cancelled, with each having a unique ID.

The 'staff' entity is needed because without it the hospital would not be functional, as staff are needed to manage the departments, schedule appointments, and treat patients. Each staff member has their own ID, first name, last name, job title, hire date, and days of work in a year. I finally I added a doctor status which was meant to be a 'BOOLEAN' but since MySQL does not support it I use 'TINYINT' (with 1 = yes, 0 = no) to easily identify which staff is actually a doctor depending on their job.

The 'patient' entity is needed since without patients none of the other entities would be existing: a hospital is for the patients. Each patient has their own ID, first name, last name, and contact details (phone number, email address, and home address).

The 'medical_history' entity is an extension to the 'patient' entity which is supposed to show past appointments and treatments for each patient. This entity will be changed in the 'Relationships, Cardinalities & Constraints' section below, in which attributes will be added.

And finally the 'appointment' entity is needed since it ties all major entities together: the patient, the staff, and the treatment (and therefore the department). It naturally has an appointment ID, an appointment date and time, and it has the primary keys of both the 'patient' and 'treatment' entities since both entities' information is needed in any appointment.

## Relationships, Cardinalities & Constraints

For the 'department' and 'treatment' entities, since there can be several treatments in several departments, a many-to-many relation has to be created between them.

For the 'appointment' entity, since it ties the 'treatment', 'staff', and 'patient' entities together, there should be the following:

- It is technically possible to have different appointments including different treatments in each appointment, but generally (and for simplicity), I'll stick to one treatment than can apply for several appointments, needing a one-to-many relation instead of a many-to-many relation.
- Since there can be several staff involved with several appointments (for example having reception and GP for one appointment), a many-to-many relation is also needed.
- And since there is usually only one patient per appointment, and said patient may have several appointments, a one-to-many relation is perfect.

The 'staff' entity needs to be connected with the 'department' entity since several staff can work in several departments without necessarily needing to have an appointment (such as janitors or receptionists). So, a many-to-many relation between the 'staff' and 'department' entity is good.

The 'patient' entity has had the 'medical_history' section taken out because it can be an entity on its own with a one-to-one relation. But since it is already possible to access a patient's medical history by going through 'patient' > 'appointment' > 'treatment' (and sorting by date and patient ID), the 'medical_history' entity can be removed.

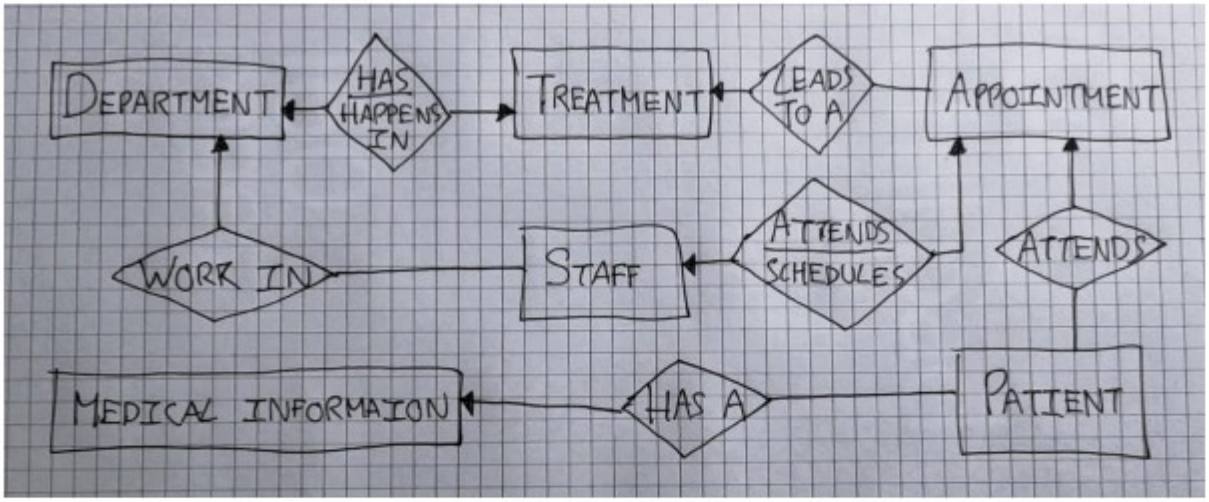
However, the 'medical_history' entity can be replaced by a 'medical_information' entity (with a one-to-one relation with the 'patient' entity) in which data can be added, containing the date of birth, blood type, height, weight, allergies, pregnancy status, medications, and medical conditions of a patient (it is possible to separate the allergies, medications, and medical conditions into separate entities, but this would be too complex and unnecessary for this project).

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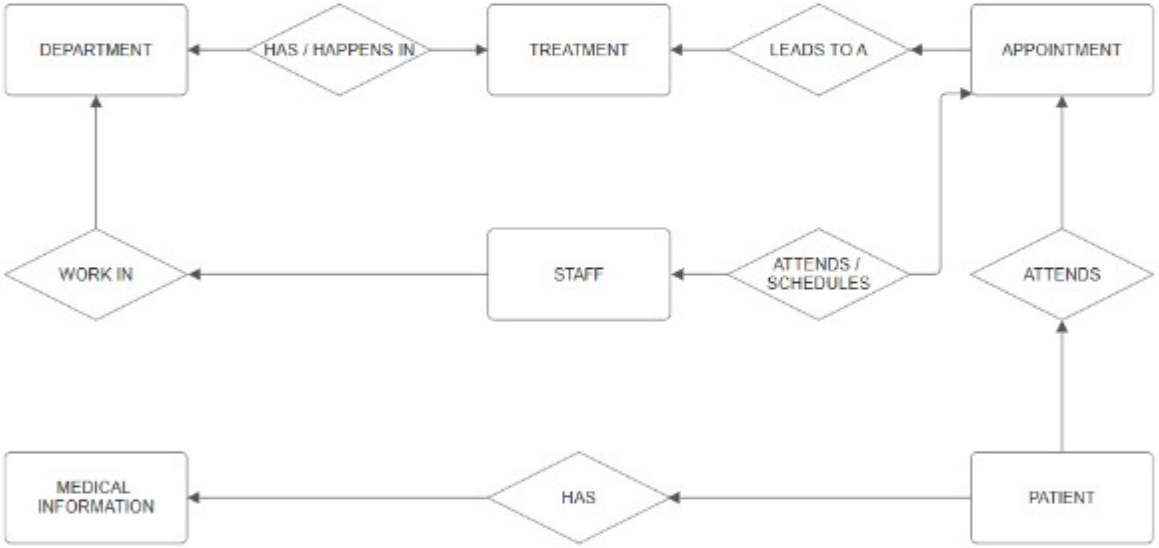
[redacted]

## Conceptual Level ERD

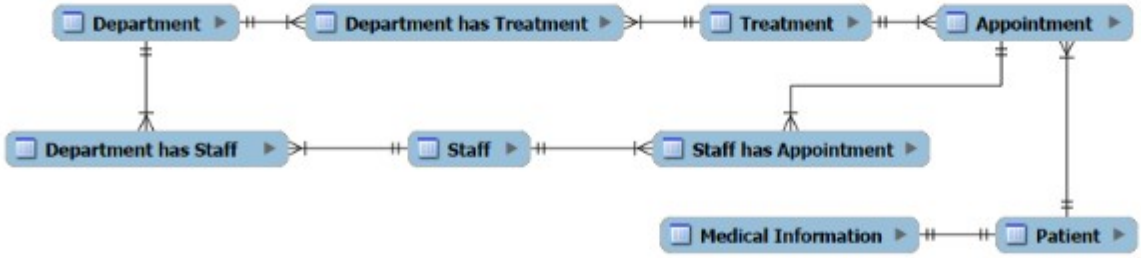
### On Paper




### On smartdraw.com



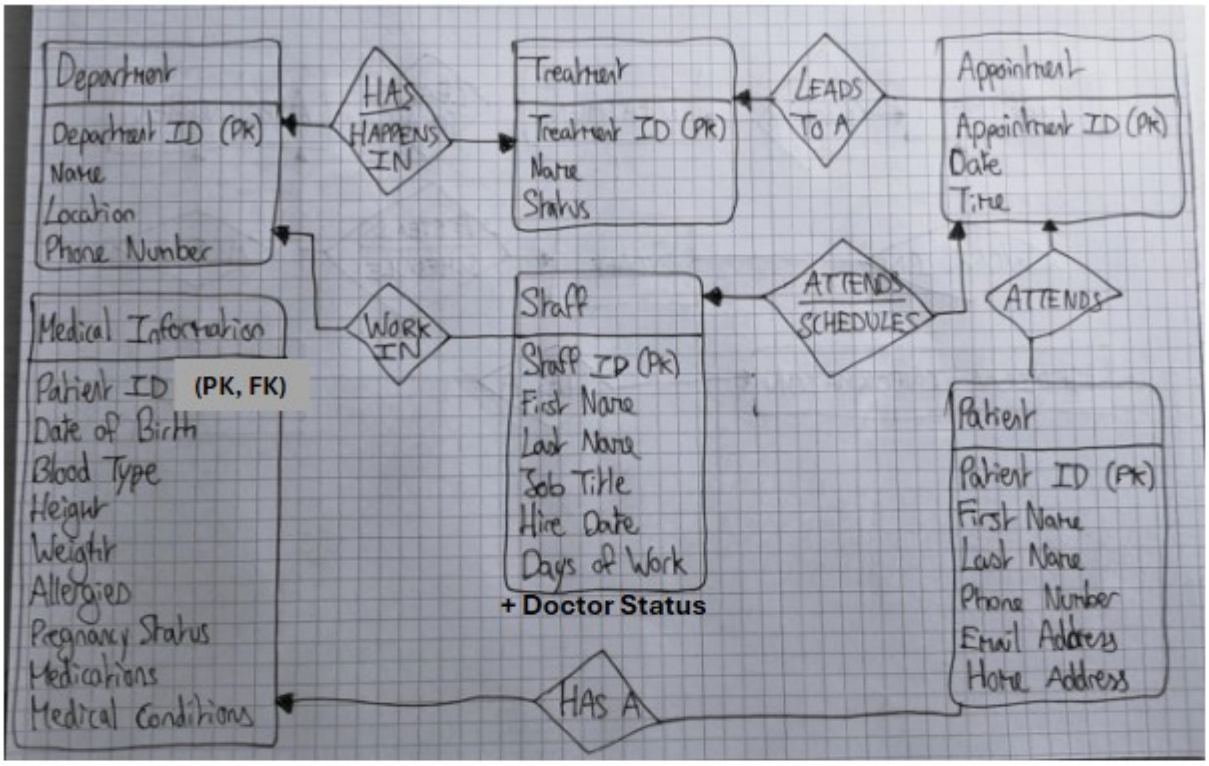

### On MySQL



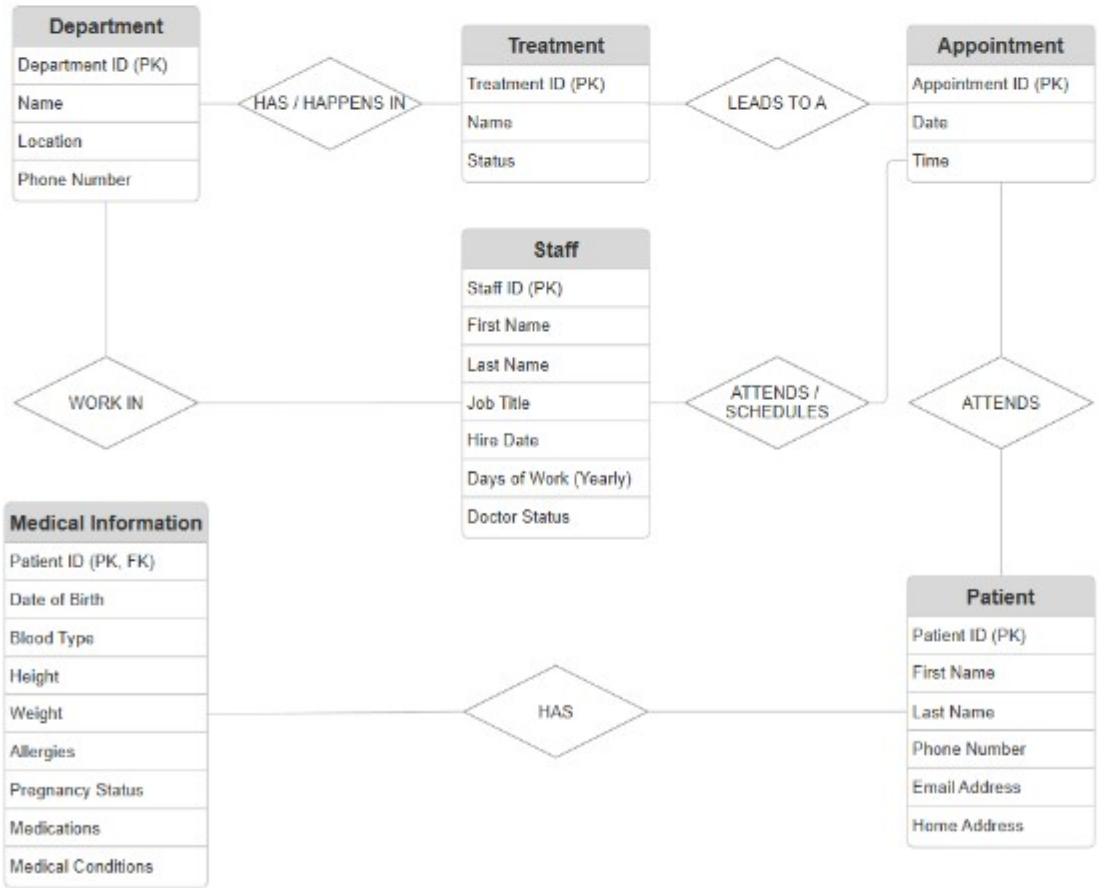
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# Logical Level ERD

## On Paper

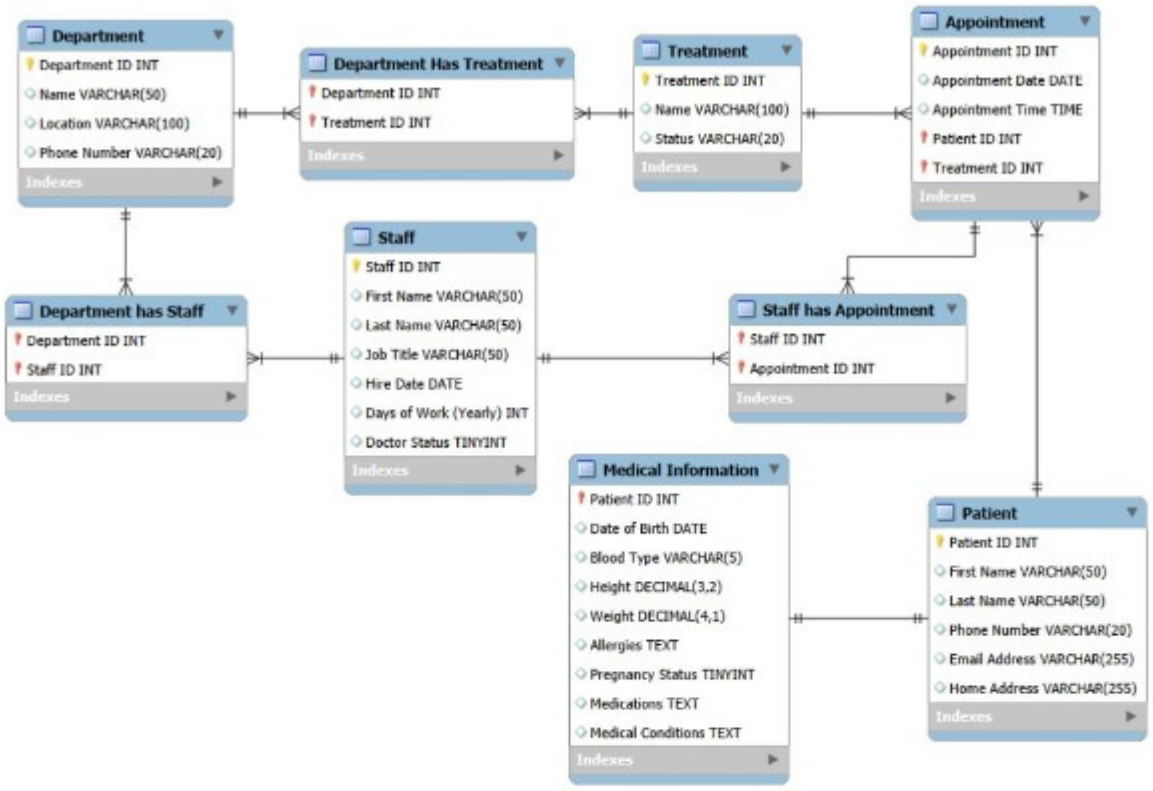



## On smartdraw.com

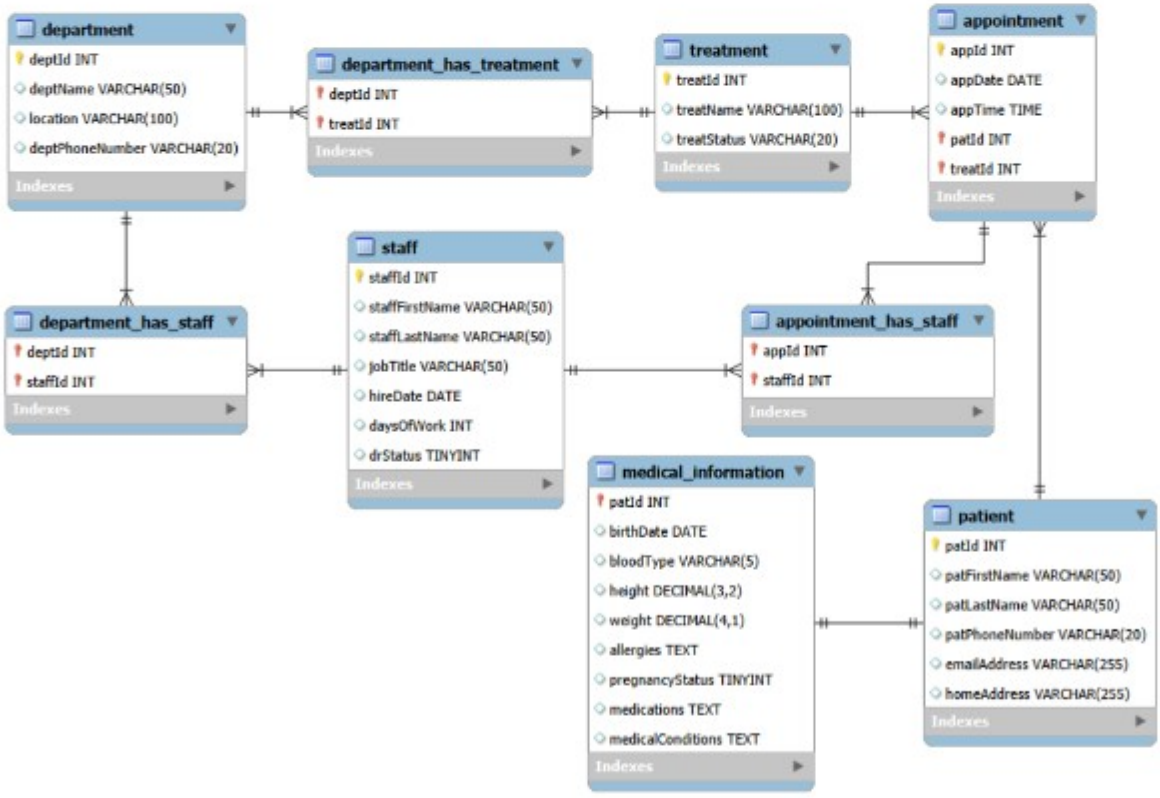


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## On MySQL (I'm obliged to add data types on MySQL)




## Physical Level ERD



## Relational Model (3NF)

- **department** (deptId (PK), deptName, location, deptPhoneNumber)
- **treatment** (treatId (PK), treatName, treatStatus)
- **staff** (staffId (PK), staffFirstName, staffLastName, jobTitle, hireDate, daysOfWork, drStatus)
- **patient** (patientId (PK), patientFirstName, patientLastName, patientPhoneNumber, emailAddress, homeAddress)
- **medical_information** (patientId (PK, FK), birthDate, bloodType, height, weight, allergies, pregnancyStatus, medications, medicalConditions)
- **appointment** (appId (PK), appDate, appTime, patientId (FK), treatId (FK))
- **department_has_treatment** (deptId (FK), treatId (FK))
- **department_has_staff** (deptId (FK), staffId (FK))
- **appointment_has_staff** (appID (FK), staffId (FK))

# Task Two – Create & Populate the Database

## Full Code

### Creating the Database

```
SET FOREIGN_KEY_CHECKS = 0; -- (I put this to allow myself to fully change primary keys of entities sharing them)

USE hospital_project;

DROP TABLE IF EXISTS appointment_has_staff;
DROP TABLE IF EXISTS department_has_staff;
DROP TABLE IF EXISTS department_has_treatment;
DROP TABLE IF EXISTS appointment;
DROP TABLE IF EXISTS medical_information;
DROP TABLE IF EXISTS patient;
DROP TABLE IF EXISTS staff;
DROP TABLE IF EXISTS treatment;
DROP TABLE IF EXISTS department;

CREATE TABLE department (
  deptId INT PRIMARY KEY,
  deptName VARCHAR(50) NOT NULL,
  location VARCHAR(100) NOT NULL,
  deptPhoneNumber VARCHAR(20)
);

CREATE TABLE treatment (
  treatId INT PRIMARY KEY,
  treatName VARCHAR(100) NOT NULL,
  treatStatus VARCHAR(20) NOT NULL
);
```

```
CREATE TABLE staff (
  staffId INT PRIMARY KEY,
  staffFirstName VARCHAR(50) NOT NULL,
  staffLastName VARCHAR(50) NOT NULL,
  jobTitle VARCHAR(50),
  hireDate DATE,
  daysOfWork INT,
  drStatus TINYINT
);

CREATE TABLE patient (
  patId INT PRIMARY KEY,
  patFirstName VARCHAR(50) NOT NULL,
  patLastName VARCHAR(50) NOT NULL,
  patPhoneNumber VARCHAR(20),
  emailAddress VARCHAR(255),
  homeAddress VARCHAR(255)
);

CREATE TABLE medical_information (
  patId INT PRIMARY KEY,
  birthDate DATE,
  bloodType VARCHAR(5),
  height DECIMAL(3,2),
  weight DECIMAL(4,1),
  allergies TEXT,
  pregnancyStatus TINYINT,
  medications TEXT,
  medicalConditions TEXT,
  FOREIGN KEY (patId) REFERENCES patient(patId)
);

CREATE TABLE appointment (
  appId INT PRIMARY KEY,
  appDate DATE NOT NULL,
  appTime TIME NOT NULL,
  patId INT NOT NULL,
  treatId INT NOT NULL,
  FOREIGN KEY (patId) REFERENCES patient(patId),
  FOREIGN KEY (treatId) REFERENCES treatment(treatId)
);

CREATE TABLE department_has_treatment (
  deptId INT,
  treatId INT,
  PRIMARY KEY (deptId, treatId),
  FOREIGN KEY (deptId) REFERENCES department(deptId),
  FOREIGN KEY (treatId) REFERENCES treatment(treatId)
```

```
);

CREATE TABLE department_has_staff (
  deptId INT,
  staffId INT,
  PRIMARY KEY (deptId, staffId),
  FOREIGN KEY (deptId) REFERENCES department(deptId),
  FOREIGN KEY (staffId) REFERENCES staff(staffId)
);

CREATE TABLE appointment_has_staff (
  appId INT,
  staffId INT,
  PRIMARY KEY (appId, staffId),
  FOREIGN KEY (appId) REFERENCES appointment(appId),
  FOREIGN KEY (staffId) REFERENCES staff(staffId)
);
```

## Populating the Database

```
INSERT INTO department (deptId, deptName, location, deptPhoneNumber) VALUES
(1, 'Cardiology', 'Block A, Level 1', '01-7650001'),
(2, 'Orthopaedics', 'Block A, Level 2', '01-7650002'),
(3, 'Orthopaedic Surgery', 'Block A, Level 3', '01-7650003'),
(4, 'Paediatrics', 'Block A, Level 4', '01-7650004'),
(5, 'General Services', 'Block B, Level 1', '01-7650005'),
(6, 'Neurology', 'Block B, Level 2', '01-7650006'),
(7, 'Radiology', 'Block C, Level 2', '01-7650007'),
(8, 'Pathology', 'Block C, Level 3', '01-7650008'),
(9, 'Pharmacy', 'Block D, Level 1', '01-7650009'),
(10, 'Intensive Care', 'Block D, Level 2', '01-7650010'),
(11, 'Emergency', 'Ground Floor', '01-7650011');

INSERT INTO treatment (treatId, treatName, treatStatus) VALUES
(1, 'Heart Surgery', 'upcoming'), (2, 'Heart Surgery', 'undergoing'), (3, 'Heart Surgery', 'completed'), (4,
'Heart Surgery', 'cancelled'),
(11, 'Knee Replacement', 'upcoming'), (12, 'Knee Replacement', 'undergoing'), (13, 'Knee
Replacement', 'completed'), (14, 'Knee Replacement', 'cancelled'),
(21, 'Routine Vaccination', 'upcoming'), (22, 'Routine Vaccination', 'undergoing'), (23, 'Routine
Vaccination', 'completed'), (24, 'Routine Vaccination', 'cancelled'),
(31, 'Diagnostic Imaging', 'upcoming'), (32, 'Diagnostic Imaging', 'undergoing'), (33, 'Diagnostic
Imaging', 'completed'), (34, 'Diagnostic Imaging', 'cancelled'),
(41, 'Spinal Tap Procedure', 'upcoming'), (42, 'Spinal Tap Procedure', 'undergoing'), (43, 'Spinal Tap
Procedure', 'completed'), (44, 'Spinal Tap Procedure', 'cancelled'),
(51, 'MRI Scan', 'upcoming'), (52, 'MRI Scan', 'undergoing'), (53, 'MRI Scan', 'completed'), (54, 'MRI
Scan', 'cancelled'),
(61, 'Biopsy Analysis', 'upcoming'), (62, 'Biopsy Analysis', 'undergoing'), (63, 'Biopsy Analysis',
'completed'), (64, 'Biopsy Analysis', 'cancelled'),
(71, 'Medication Dispensing', 'upcoming'), (72, 'Medication Dispensing', 'undergoing'), (73, 'Medication
Dispensing', 'completed'), (74, 'Medication Dispensing', 'cancelled'),
```

(81, 'Mechanical Ventilation', 'upcoming'), (82, 'Mechanical Ventilation', 'undergoing'), (83, 'Mechanical Ventilation', 'completed'), (84, 'Mechanical Ventilation', 'cancelled'),
(91, 'Trauma Stabilization', 'upcoming'), (92, 'Trauma Stabilization', 'undergoing'), (93, 'Trauma Stabilization', 'completed'), (94, 'Trauma Stabilization', 'cancelled');

INSERT INTO staff (staffId, staffFirstName, staffLastName, jobTitle, hireDate, daysOfWork, drStatus) VALUES
(1, 'Robert', 'Downs', 'CEO', '2010-01-01', 200, 0),
(2, 'Kavin', 'Smith', 'Cardiologist', '2015-05-20', 240, 1),
(3, 'Serena', 'Khan', 'Orthopaedic Surgeon', '2018-01-15', 235, 1),
(4, 'Jasmine', 'Lee', 'Senior Paediatric Nurse', '2020-11-01', 250, 0),
(5, 'Marcus', 'Wright', 'Office Worker', '2017-03-10', 260, 1),
(6, 'Olivia', 'Chen', 'Neurophysiologist', '2019-07-25', 245, 0),
(7, 'Rohan', 'Patel', 'MRI Technologist', '2016-09-01', 220, 0),
(8, 'Amelia', 'Scott', 'Laboratory Scientist', '2014-04-05', 230, 0),
(9, 'Ben', 'Carter', 'Hospital Pharmacist', '2021-02-14', 240, 0),
(10, 'Chloe', 'Hayes', 'ICU Lead Clinician', '2022-08-01', 255, 1),
(11, 'David', 'Evans', 'Emergency Medicine Ragistrar', '2023-01-20', 260, 1);

INSERT INTO patient (patId, patFirstName, patLastName, patPhoneNumber, emailAddress, homeAddress) VALUES
(1298, 'Alice', 'Smith', '02-8760001', 'alice.s@email.com', '1 High Street, London'),
(2632, 'Robert', 'Jones', '02-8760002', 'robert.j@email.com', '15 Oak Lane, Birmingham'),
(3456, 'Laura', 'Gomez', '02-8760003', 'laura.g@email.com', '42 Elm Road, Manchester'),
(4127, 'Michael', 'Brown', '02-8760004', 'michael.b@email.com', '8 The Avenue, Bristol'),
(5612, 'John', 'Bob', '02-876005', 'john.b@email.com', '12 Old Town, Belfast'),
(6121, 'Sarah', 'Wilson', '02-8760006', 'sarah.w@email.com', '29 Park Row, Leeds'),
(7192, 'James', 'Miller', '02-8760007', 'james.m@email.com', '6 Willow Crescent, Edinburgh'),
(8120, 'Emily', 'Clark', '02-8760008', 'emily.c@email.com', '10 Forest View, Glasgow'),
(9310, 'Daniel', 'Walker', '02-8760009', 'daniel.w@email.com', '33 River Lane, Dublin'),
(1247, 'Jessica', 'Hall', '02-8760010', 'jessica.h@email.com', '5 New Road, Cardiff');

INSERT INTO medical_information (patId, birthDate, bloodType, height, weight, allergies, pregnancyStatus, medications, medicalConditions) VALUES
(1298, '1985-06-15', 'O+', 1.68, 65.2, 'Penicillin', 0, 'Metoprolol', 'Hypertension'),
(2632, '1972-11-20', 'A-', 1.80, 88.5, 'None', 0, 'Aspirin', 'Osteoarthritis'),
(3456, '2019-03-01', 'B+', 0.95, 14.7, 'Dairy', 1, 'Vitamins', 'Asthma'),
(4127, '1999-09-09', 'AB+', 1.75, 75.0, 'Pollen', 0, 'None', 'Diabetes'),
(5612, '1980-12-30', 'AB-', 1.78, 84.0, 'Pollen', 0, 'Ibuprofen', 'Mild Arthritis'),
(6121, '1995-04-12', 'O-', 1.70, 71.3, 'Nuts', 0, 'Vitamins', 'Asthma'),
(7192, '1965-01-28', 'A+', 1.90, 95.9, 'Latex', 0, 'Statins', 'High Cholesterol, Diabetes'),
(8120, '1988-10-05', 'B-', 1.63, 62.1, 'None', 0, 'Sertraline', 'Migraines'),
(9310, '1990-07-07', 'O+', 1.82, 80.4, 'Shellfish', 0, 'None', 'Concussion History'),
(1247, '2005-02-22', 'A-', 1.55, 55.6, 'Cats', 0, 'None', 'Anxiety, Diabetes');

INSERT INTO appointment (appId, appDate, appTime, patId, treatId) VALUES
(1, '2025-10-10', '10:00:00', 1298, 2),
(2, '2025-09-05', '14:30:00', 2632, 11),
(3, '2025-09-15', '09:00:00', 3456, 22),



```
(4, '2025-12-01', '19:45:00', 4127, 34),
(5, '2025-12-20', '11:30:00', 5612, 43),
(6, '2025-12-25', '16:00:00', 6121, 52),
(7, '2025-08-10', '08:30:00', 7192, 63),
(8, '2025-09-01', '13:00:00', 8120, 71),
(9, '2025-10-05', '10:30:00', 9310, 84),
(10, '2025-12-02', '20:30:00', 1247, 91);

INSERT INTO department_has_treatment (deptId, treatId) VALUES
(1, 1), (1, 2), (1, 3), (1, 4),
(2, 11), (2, 12), (2, 13), (2, 14),
(4, 21), (4, 22), (4, 23), (4, 24),
(5, 31), (5, 32), (5, 33), (5, 34),
(6, 41), (6, 42), (6, 43), (6, 44),
(7, 51), (7, 52), (7, 53), (7, 54),
(8, 61), (8, 62), (8, 63), (8, 64),
(9, 71), (9, 72), (9, 73), (9, 74),
(10, 81), (10, 82), (10, 83), (10, 84),
(11, 91), (11, 92), (11, 93), (11, 94);

INSERT INTO department_has_staff (deptId, staffId) VALUES
(1, 2),
(3, 3),
(4, 4),
(5, 5),
(6, 6),
(7, 7),
(8, 8),
(9, 9),
(10, 10),
(11, 11);

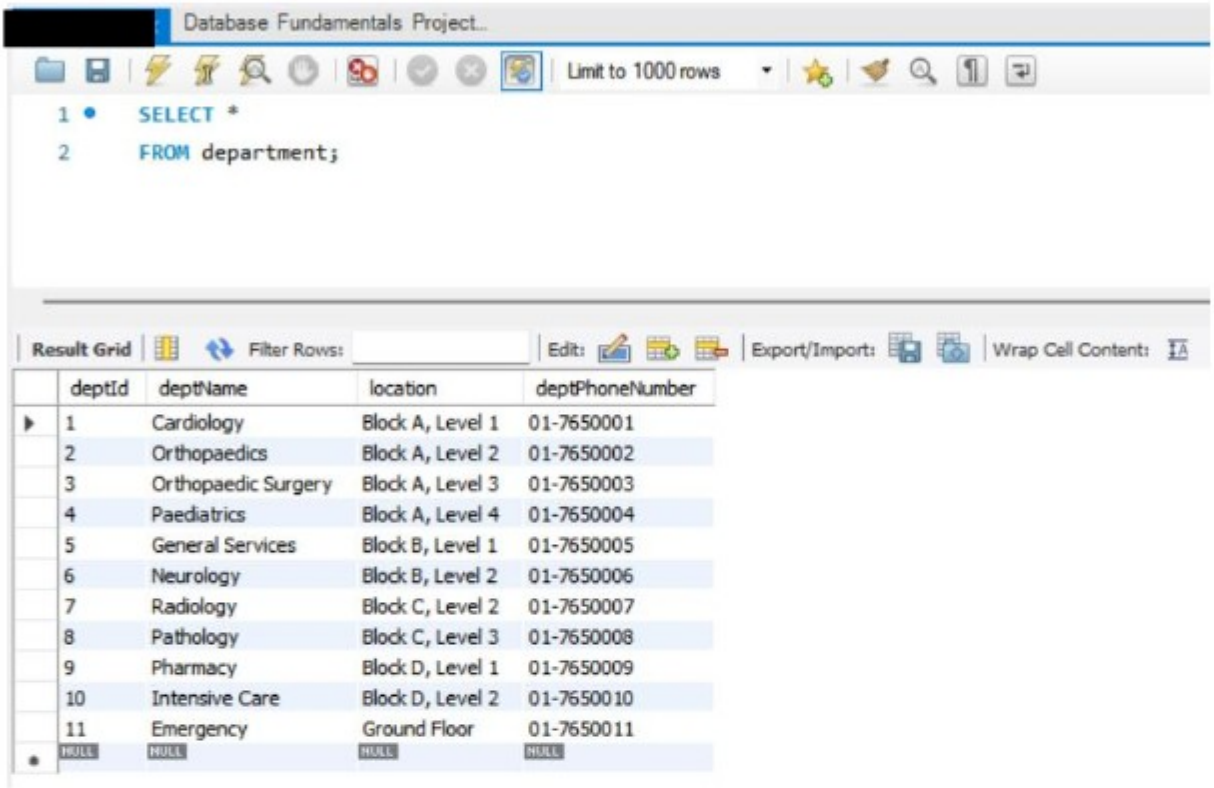
INSERT INTO appointment_has_staff (appId, staffId) VALUES
(1, 2),
(2, 3),
(3, 4),
(4, 5),
(5, 6),
(6, 7),
(7, 8),
(8, 9),
(9, 10),
(10, 11);

SET FOREIGN_KEY_CHECKS = 1; -- (setting it back to 1 to avoid breaking anything later on)
```

# Task Three – Query the Dataset

## Show details of all medical departments of hospital.

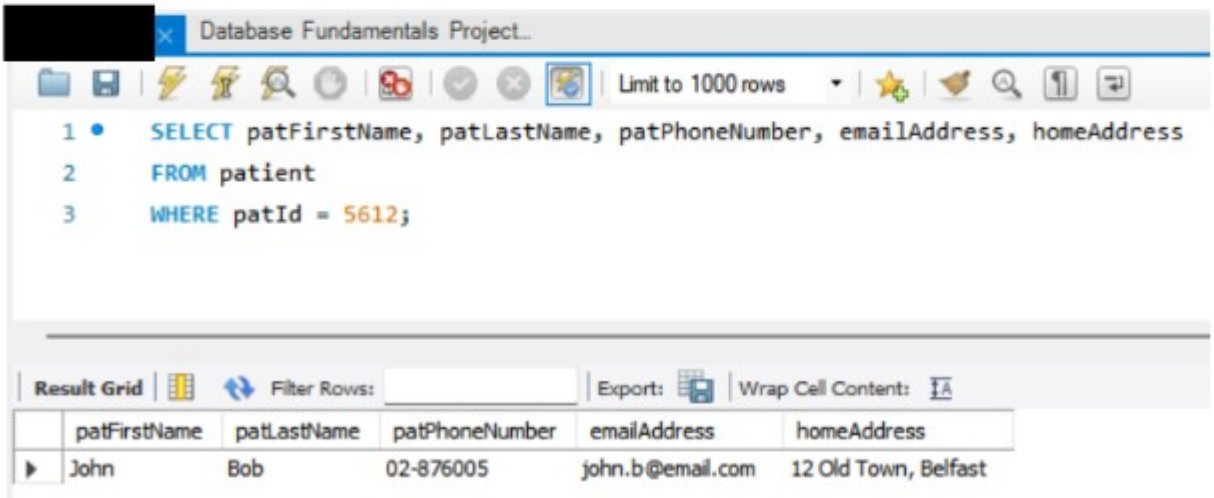
SELECT *
FROM department;




| deptId | deptName | location | deptPhoneNumber |
|---|---|---|---|
| 1 | Cardiology | Block A, Level 1 | 01-7650001 |
| 2 | Orthopaedics | Block A, Level 2 | 01-7650002 |
| 3 | Orthopaedic Surgery | Block A, Level 3 | 01-7650003 |
| 4 | Paediatrics | Block A, Level 4 | 01-7650004 |
| 5 | General Services | Block B, Level 1 | 01-7650005 |
| 6 | Neurology | Block B, Level 2 | 01-7650006 |
| 7 | Radiology | Block C, Level 2 | 01-7650007 |
| 8 | Pathology | Block C, Level 3 | 01-7650008 |
| 9 | Pharmacy | Block D, Level 1 | 01-7650009 |
| 10 | Intensive Care | Block D, Level 2 | 01-7650010 |
| 11 | Emergency | Ground Floor | 01-7650011 |
| NULL | NULL | NULL | NULL |

## Find the contact details of patient “John Bob” whose patient ID is 5612.

SELECT patFirstName, patLastName, patPhoneNumber, emailAddress, homeAddress
FROM patient
WHERE patId = 5612;

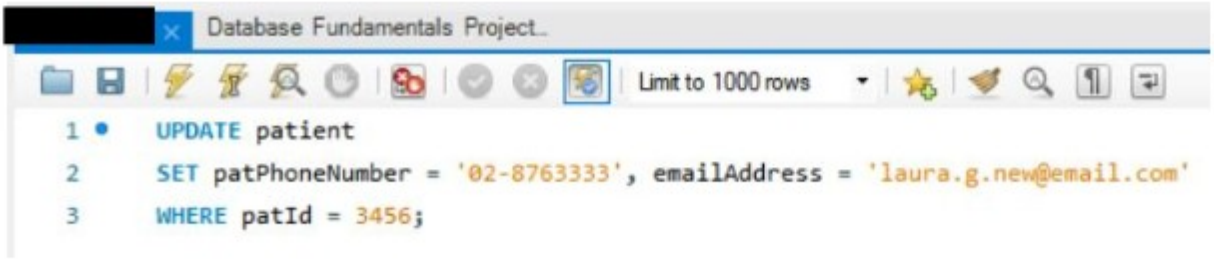



| patFirstName | patLastName | patPhoneNumber | emailAddress | homeAddress |
|---|---|---|---|---|
| John | Bob | 02-876005 | john.b@email.com | 12 Old Town, Belfast |

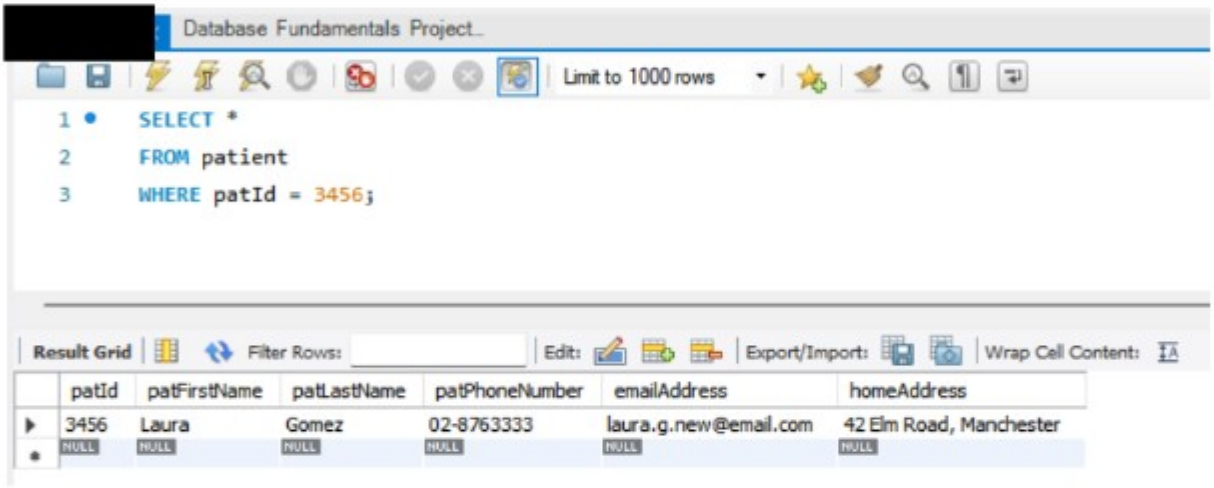
## Update the patient ID number 3456 contact details.

UPDATE patient

SET patPhoneNumber = '02-8763333', emailAddress = 'laura.g.new@email.com'
WHERE patId = 3456;

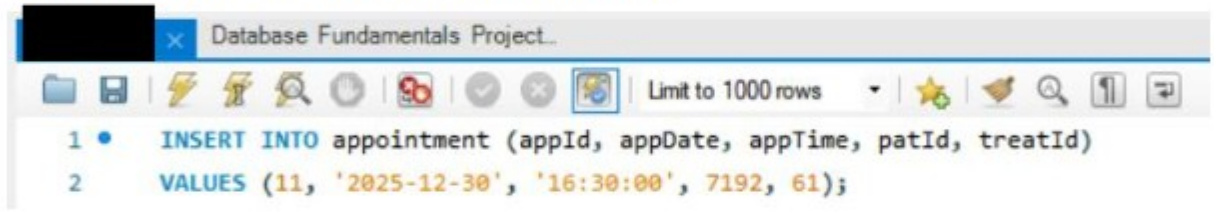



## Proof

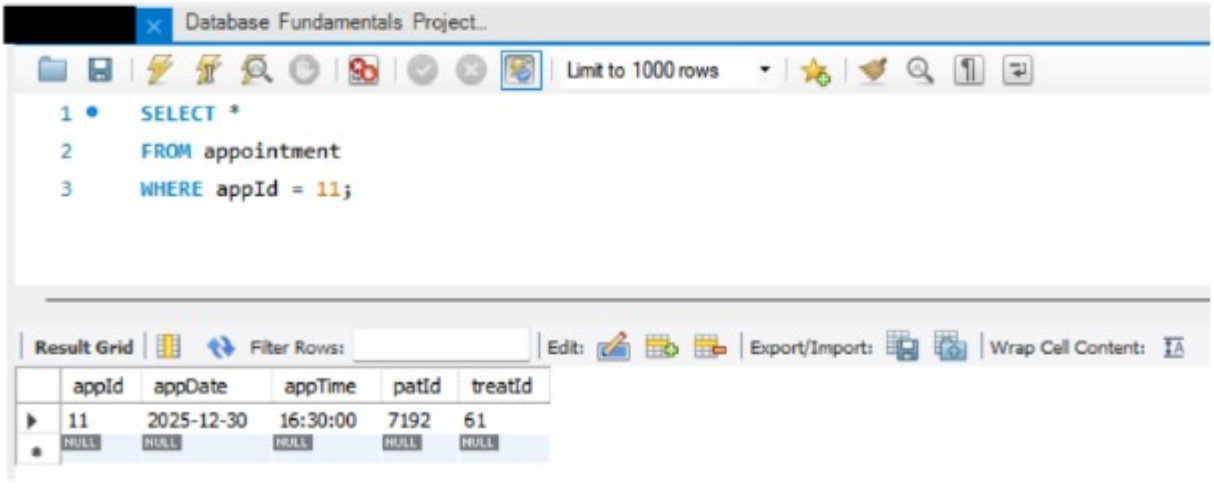



## Schedule a new medical appointment.

INSERT INTO appointment (appId, appDate, appTime, patId, treatId)
VALUES (11, '2025-12-30', '16:30:00', 7192, 61);

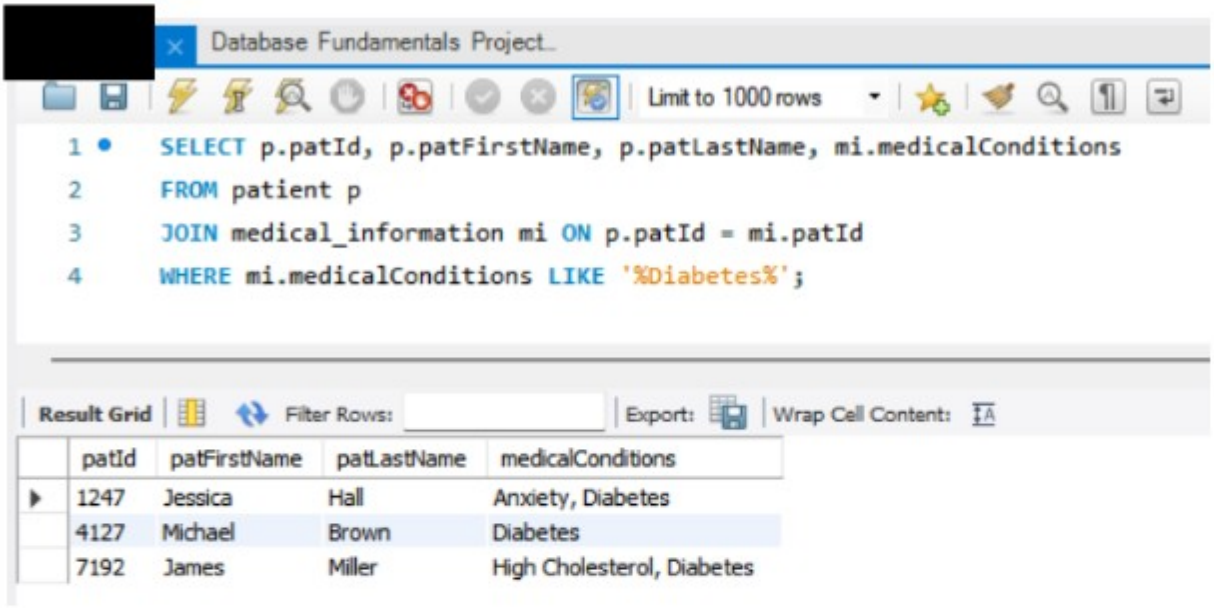



## Proof



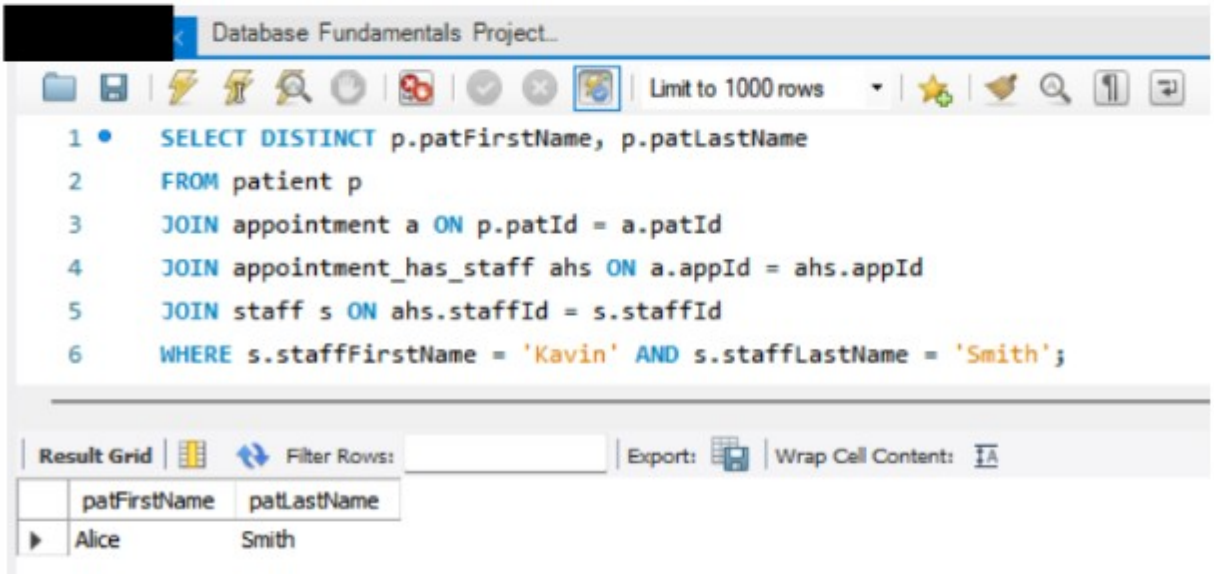
## Find details of all patients who have Diabetes.

```
SELECT p.patId, p.patFirstName, p.patLastName, mi.medicalConditions
FROM patient p
JOIN medical_information mi ON p.patId = mi.patId
WHERE mi.medicalConditions LIKE '%Diabetes%';
```



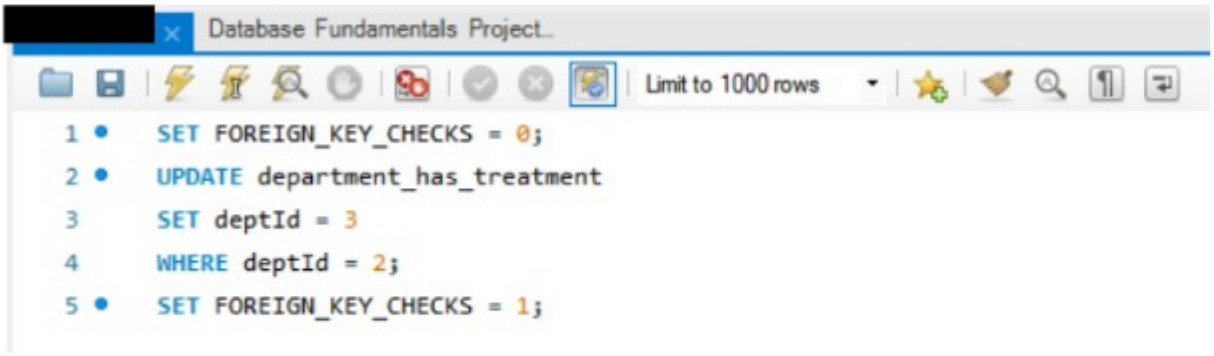

## Display the first name and surname of all patients who are under the care of Dr. Kavin Smith.

```
SELECT DISTINCT p.patFirstName, p.patLastName
FROM patient p
JOIN appointment a ON p.patId = a.patId
JOIN appointment_has_staff ahs ON a.appId = ahs.appId
JOIN staff s ON ahs.staffId = s.staffId
WHERE s.staffFirstName = 'Kavin' AND s.staffLastName = 'Smith';
```

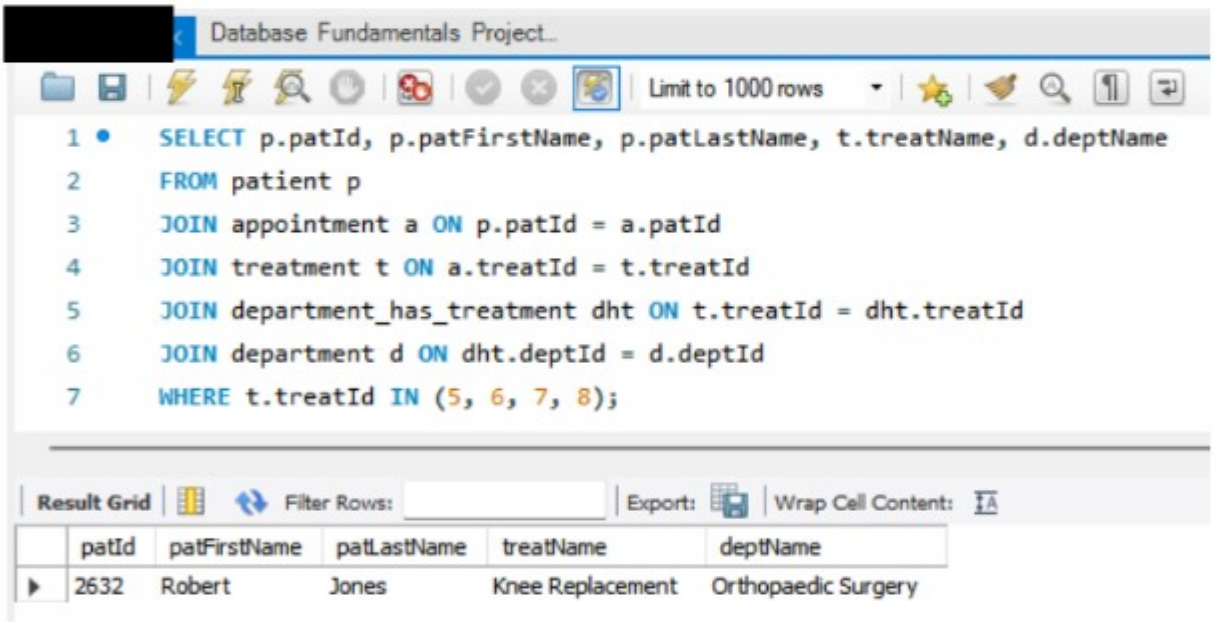


## Change all patients in Orthopaedics department to Orthopaedic Surgery department.

```
SET FOREIGN_KEY_CHECKS = 0;
UPDATE department_has_treatment
SET deptId = 3
WHERE deptId = 2;
SET FOREIGN_KEY_CHECKS = 1;
```




Proof




## Display the first name, surname, and hire date of all doctors sorted in descending order by hire date.

```
SELECT staffFirstName, staffLastName, hireDate
FROM staff
WHERE drStatus = 1
ORDER BY hireDate DESC;
```

Maël Conejos Crusson

[redacted]

Database Fundamentals Project...

Limit to 1000 rows

```
1 •  SELECT staffFirstName, staffLastName, hireDate
2    FROM staff
3    WHERE drStatus = 1
4    ORDER BY hireDate DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| staffFirstName | staffLastName | hireDate |
|---|---|---|
| David | Evans | 2023-01-20 |
| Chloe | Hayes | 2022-08-01 |
| Serena | Khan | 2018-01-15 |
| Marcus | Wright | 2017-03-10 |
| Kavin | Smith | 2015-05-20 |

## Count how many patients are undergoing 'Heart Surgery' treatment.

```
SELECT COUNT(DISTINCT p.patId) AS patientsUndergoingHeartSurgery
FROM patient p
JOIN appointment a ON p.patId = a.patId
JOIN treatment t ON a.treatId = t.treatId
WHERE t.treatName = 'Heart Surgery' AND t.treatStatus = 'undergoing';
```

[redacted]

Database Fundamentals Project...

Limit to 1000 rows

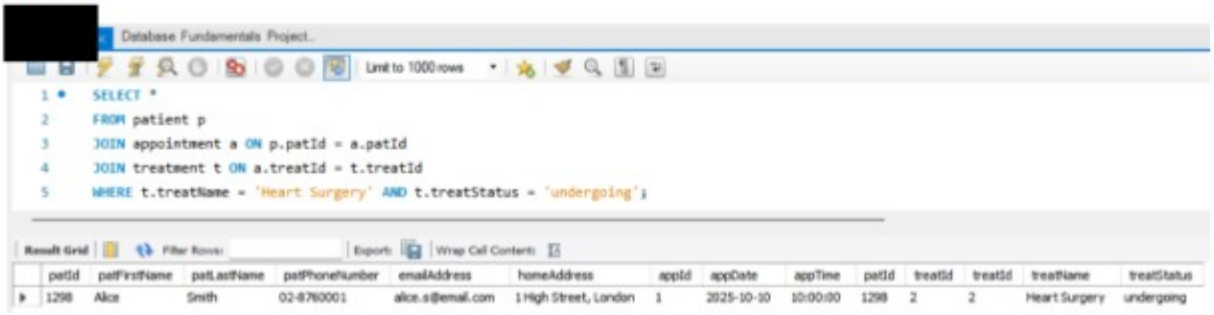
```
1 •  SELECT COUNT(DISTINCT p.patId) AS patientsUndergoingHeartSurgery
2    FROM patient p
3    JOIN appointment a ON p.patId = a.patId
4    JOIN treatment t ON a.treatId = t.treatId
5    WHERE t.treatName = 'Heart Surgery' AND t.treatStatus = 'undergoing';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| patientsUndergoingHeartSurgery |
|---|
| 1 |

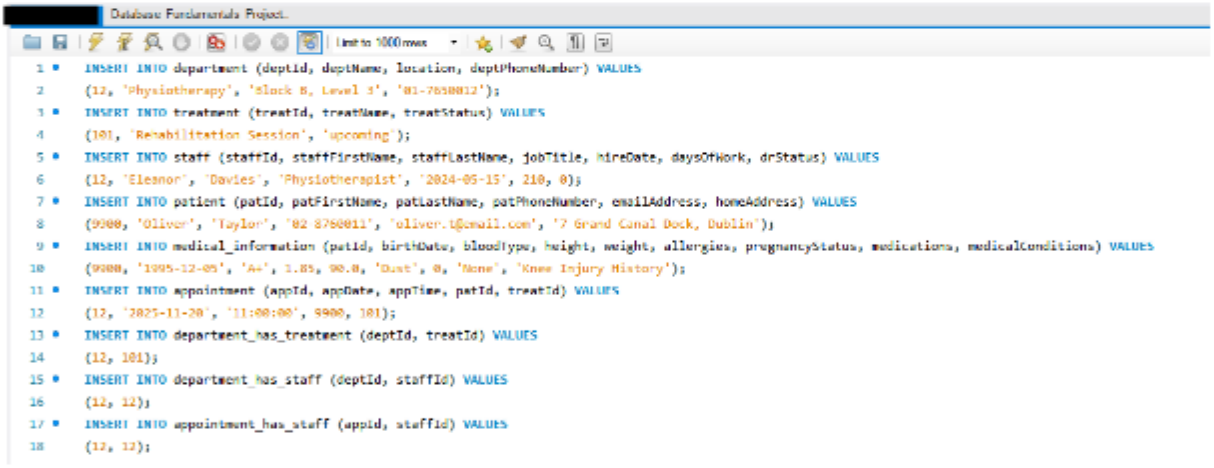
## Proof

[redacted]

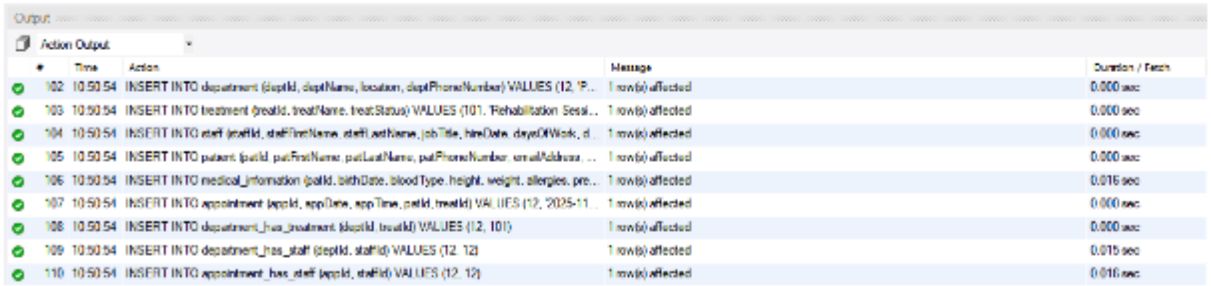


## Add one new record to each table.

```
INSERT INTO department (deptId, deptName, location, deptPhoneNumber) VALUES
(12, 'Physiotherapy', 'Block B, Level 3', '01-7650012');
INSERT INTO treatment (treatId, treatName, treatStatus) VALUES
(101, 'Rehabilitation Session', 'upcoming');
INSERT INTO staff (staffId, staffFirstName, staffLastName, jobTitle, hireDate, daysOfWork, drStatus)
VALUES
(12, 'Eleanor', 'Davies', 'Physiotherapist', '2024-05-15', 210, 0);
INSERT INTO patient (patId, patFirstName, patLastName, patPhoneNumber, emailAddress,
homeAddress) VALUES
(9900, 'Oliver', 'Taylor', '02-8760011', 'oliver.t@email.com', '7 Grand Canal Dock, Dublin');
INSERT INTO medical_information (patId, birthDate, bloodType, height, weight, allergies,
pregnancyStatus, medications, medicalConditions) VALUES
(9900, '1995-12-05', 'A+', 1.85, 90.0, 'Dust', 0, 'None', 'Knee Injury History');
INSERT INTO appointment (appId, appDate, appTime, patId, treatId) VALUES
(12, '2025-11-20', '11:00:00', 9900, 101);
INSERT INTO department_has_treatment (deptId, treatId) VALUES
(12, 101);
INSERT INTO department_has_staff (deptId, staffId) VALUES
(12, 12);
INSERT INTO appointment_has_staff (appId, staffId) VALUES
(12, 12);
```



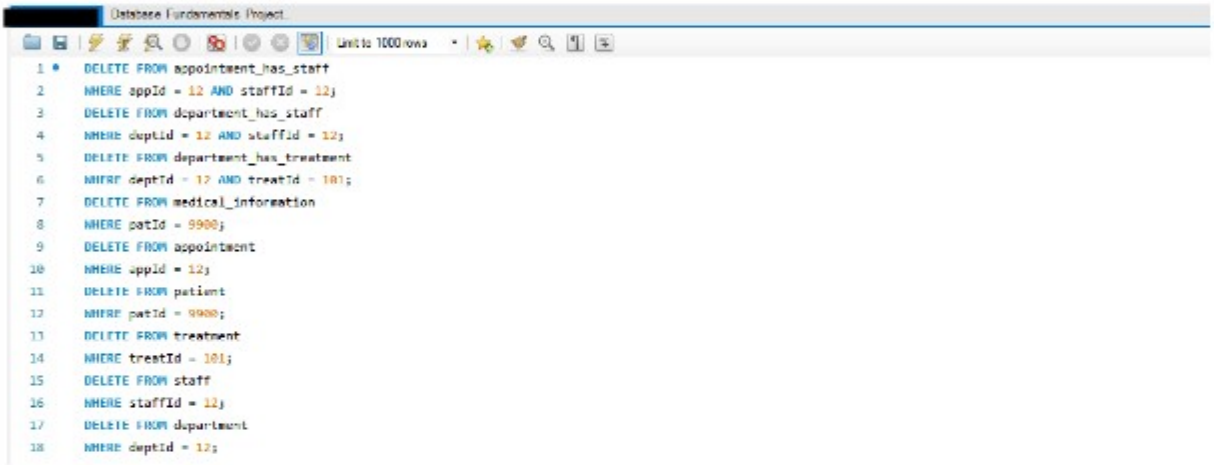

## Proof



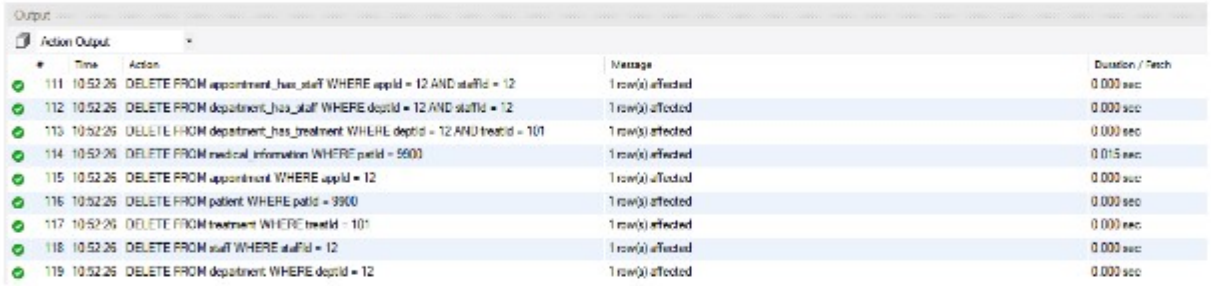

## Delete one record from all tables.

```
DELETE FROM appointment_has_staff
WHERE appId = 12 AND staffId = 12;
```

```
DELETE FROM department_has_staff
WHERE deptId = 12 AND staffId = 12;
DELETE FROM department_has_treatment
WHERE deptId = 12 AND treatId = 101;
DELETE FROM medical_information
WHERE patId = 9900;
DELETE FROM appointment
WHERE appId = 12;
DELETE FROM patient
WHERE patId = 9900;
DELETE FROM treatment
WHERE treatId = 101;
DELETE FROM staff
WHERE staffId = 12;
DELETE FROM department
WHERE deptId = 12;
```




## Proof




## Find the total number of days off for all staff, order this by least days off.

```
SELECT staffFirstName, staffLastName, (365 - daysOfWork) AS totalDaysOff
FROM staff
ORDER BY TotalDaysOff ASC;
```

[redacted]

Database Fundamentals Project...

Limit to 1000 rows

```
1 •  SELECT staffFirstName, staffLastName, (365 - daysOfWork) AS totalDaysOff
2    FROM staff
3    ORDER BY TotalDaysOff ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| staffFirstName | staffLastName | totalDaysOff |
|---|---|---|
| Marcus | Wright | 105 |
| David | Evans | 105 |
| Chloe | Hayes | 110 |
| Jasmine | Lee | 115 |
| Olivia | Chen | 120 |
| Kavin | Smith | 125 |
| Ben | Carter | 125 |
| Serena | Khan | 130 |
| Amelia | Scott | 135 |
| Rohan | Patel | 145 |
| Robert | Downs | 165 |

## Change the role of all staff who have a job title 'Office Worker' to 'Administrator'.

UPDATE staff
SET jobTitle = 'Administrator'
WHERE jobTitle = 'Office Worker';

[redacted]

Database Fundamentals Project...

Limit to 1000 rows

```
1 •  UPDATE staff
2    SET jobTitle = 'Administrator'
3    WHERE jobTitle = 'Office Worker';
```

## Proof

[redacted]

Database Fundamentals Project...

Limit to 1000 rows

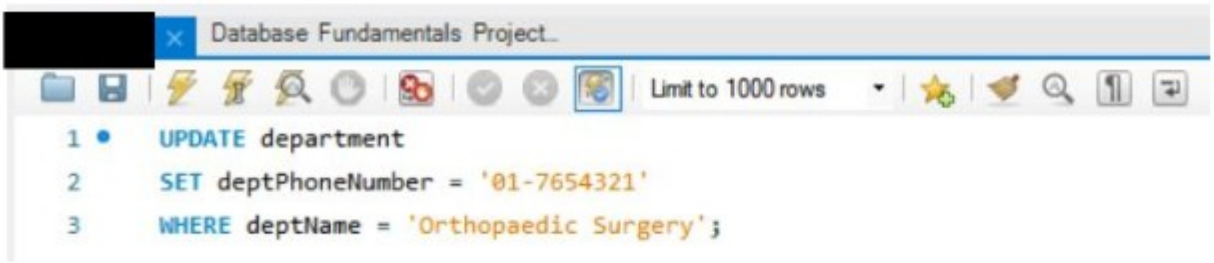
```
1 •  SELECT *
2    FROM staff
3    WHERE jobTitle = 'Administrator';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

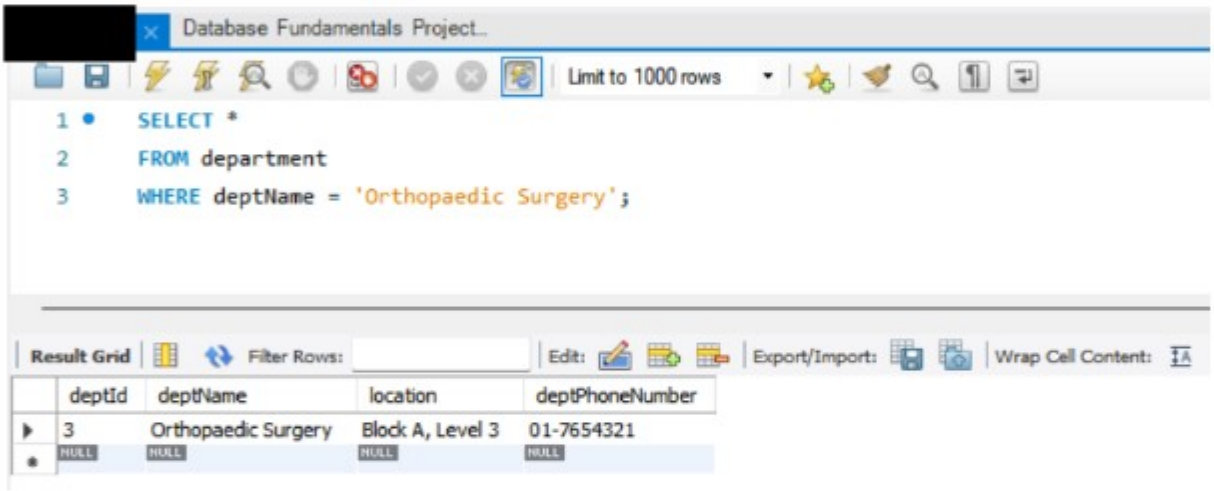
| staffId | staffFirstName | staffLastName | jobTitle | hireDate | daysOfWork | drStatus |
|---|---|---|---|---|---|---|
| 5 | Marcus | Wright | Administrator | 2017-03-10 | 260 | 1 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

## Set the Orthopaedic Surgery department phone number to be 01-7654321.

```
UPDATE department
SET deptPhoneNumber = '01-7654321'
WHERE deptName = 'Orthopaedic Surgery';
```

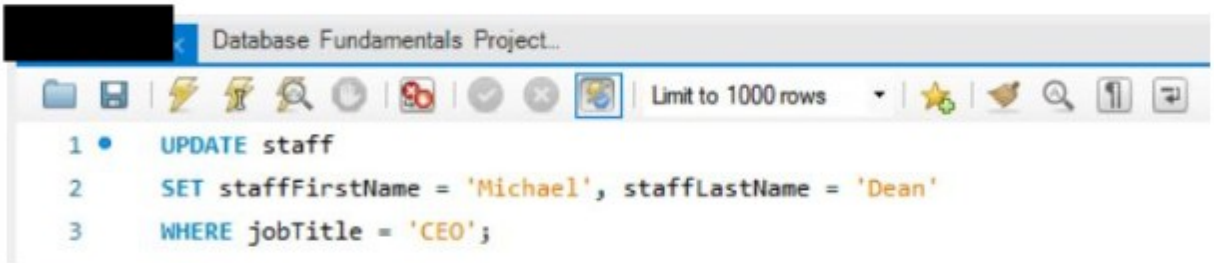



Proof



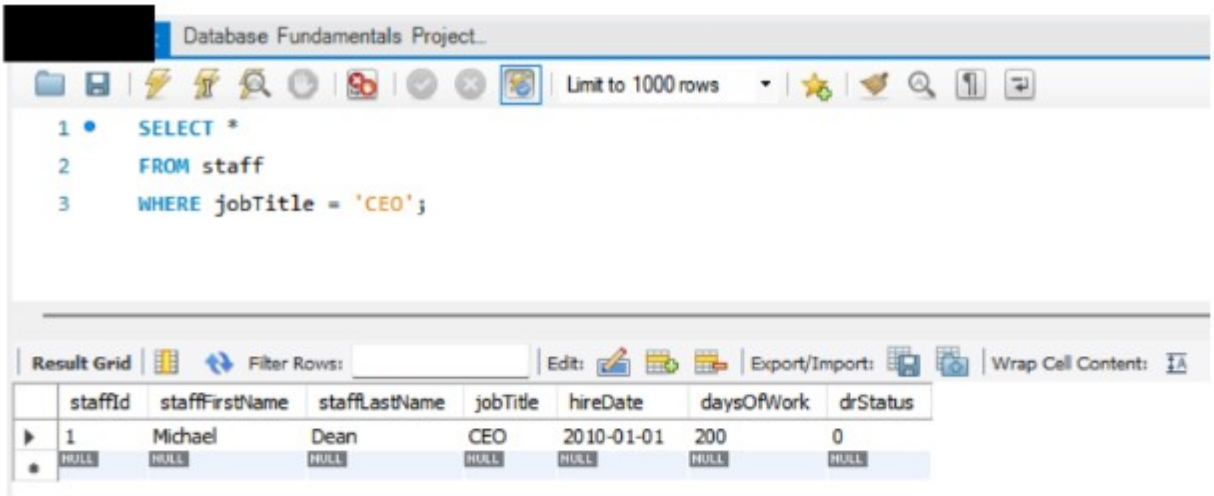

## Set the chief executive officer (CEO) of the hospital to be called Michael Dean.

```
UPDATE staff
SET staffFirstName = 'Michael', staffLastName = 'Dean'
WHERE jobTitle = 'CEO';
```



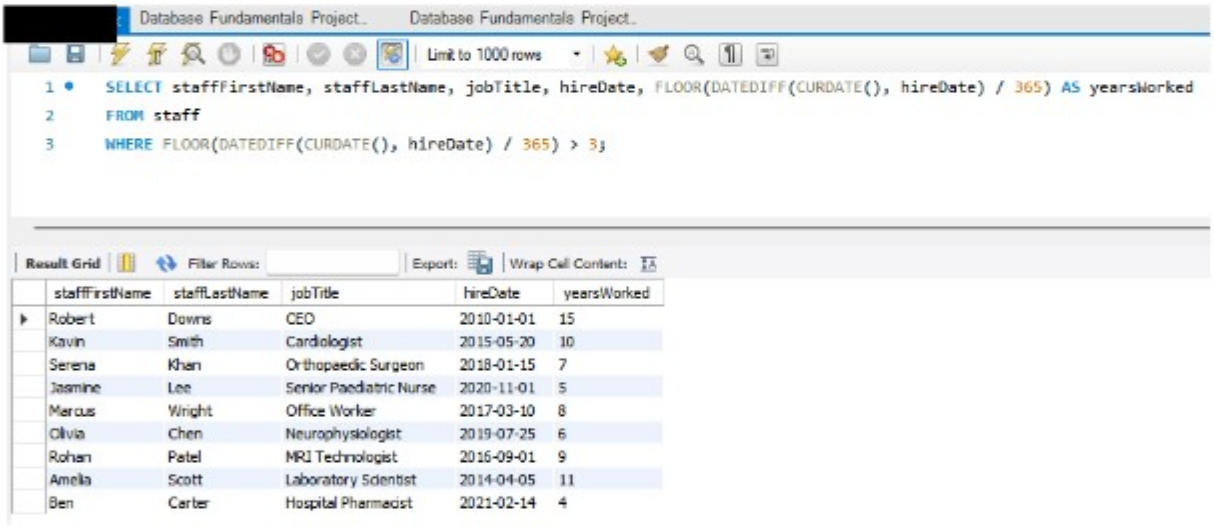
Proof

[redacted]




## Show all staff members who have been working there for longer than 3 years.

SELECT staffFirstName, staffLastName, jobTitle, hireDate, FLOOR(DATEDIFF(CURDATE(), hireDate) / 365) AS yearsWorked
FROM staff
WHERE FLOOR(DATEDIFF(CURDATE(), hireDate) / 365) > 3;

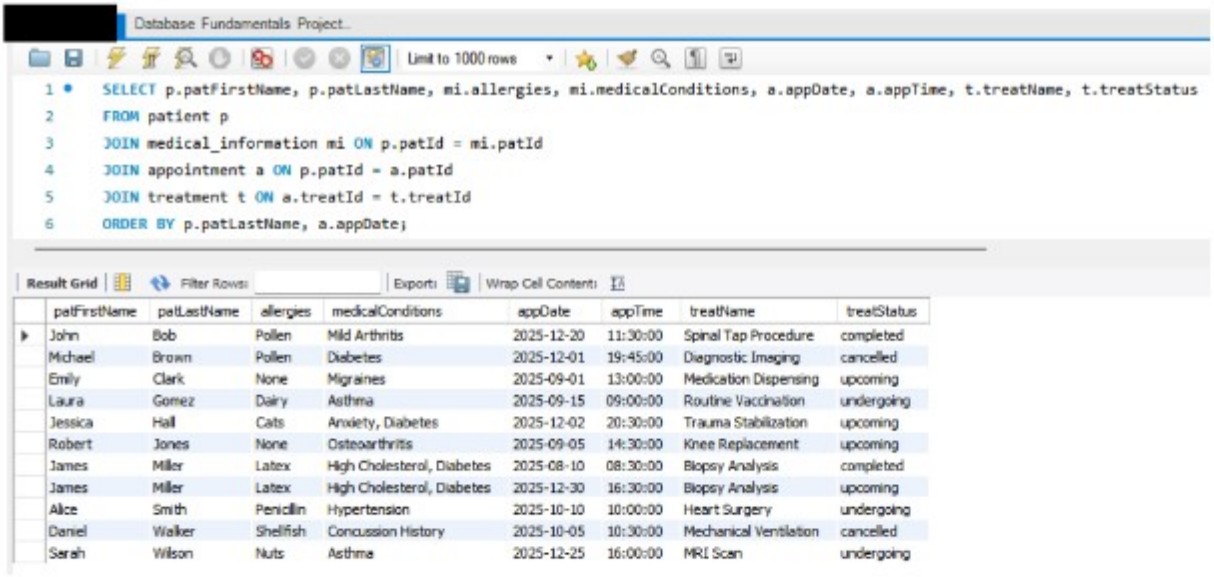



## Show the detail of all patient names, medical history, appointment dates, times, and treatment.

SELECT p.patFirstName, p.patLastName, mi.allergies, mi.medicalConditions, a.appDate, a.appTime, t.treatName, t.treatStatus
FROM patient p
JOIN medical_information mi ON p.patId = mi.patId
JOIN appointment a ON p.patId = a.patId
JOIN treatment t ON a.treatId = t.treatId
ORDER BY p.patLastName, a.appDate;

Maël Conejos Crusson

[redacted]

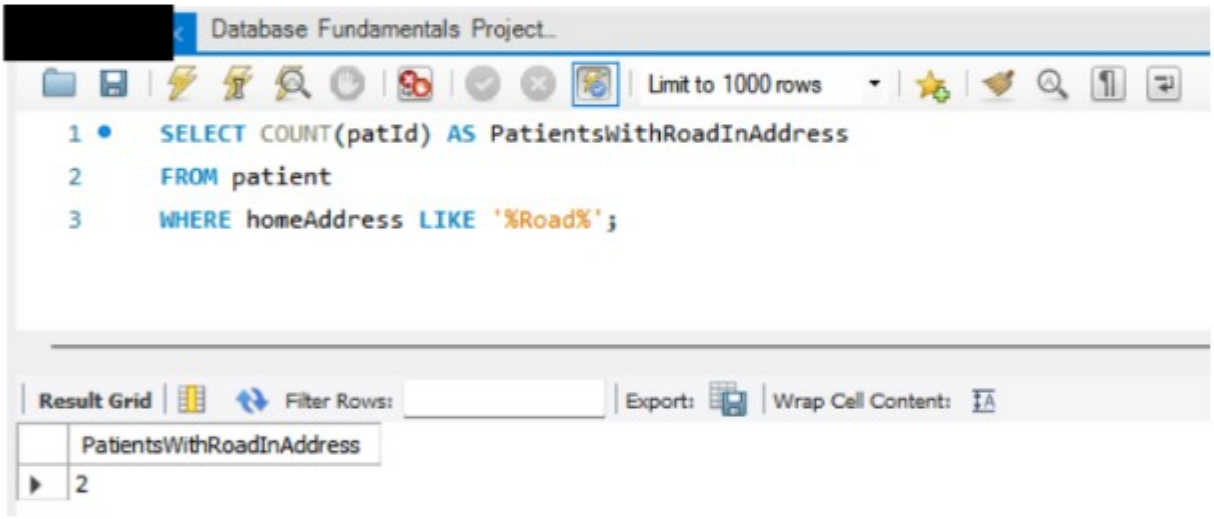



| patFirstName | patLastName | allergies | medicalConditions | appDate | appTime | treatName | treatStatus |
|---|---|---|---|---|---|---|---|
| John | Bob | Pollen | Mild Arthritis | 2025-12-20 | 11:30:00 | Spinal Tap Procedure | completed |
| Michael | Brown | Pollen | Diabetes | 2025-12-01 | 19:45:00 | Diagnostic Imaging | cancelled |
| Emily | Clark | None | Migraines | 2025-09-01 | 13:00:00 | Medication Dispensing | upcoming |
| Laura | Gomez | Dairy | Asthma | 2025-09-15 | 09:00:00 | Routine Vaccination | undergoing |
| Jessica | Hall | Cats | Anxiety, Diabetes | 2025-12-02 | 20:30:00 | Trauma Stabilization | upcoming |
| Robert | Jones | None | Osteoarthritis | 2025-09-05 | 14:30:00 | Knee Replacement | upcoming |
| James | Miller | Latex | High Cholesterol, Diabetes | 2025-08-10 | 08:30:00 | Biopsy Analysis | completed |
| James | Miller | Latex | High Cholesterol, Diabetes | 2025-12-30 | 16:30:00 | Biopsy Analysis | upcoming |
| Alice | Smith | Penicillin | Hypertension | 2025-10-10 | 10:00:00 | Heart Surgery | undergoing |
| Daniel | Walker | Shellfish | Concussion History | 2025-10-05 | 10:30:00 | Mechanical Ventilation | cancelled |
| Sarah | Wilson | Nuts | Asthma | 2025-12-25 | 16:00:00 | MRI Scan | undergoing |

## Identify how many patients have the word 'road' in their address.

SELECT COUNT(patId) AS PatientsWithRoadInAddress
FROM patient
WHERE homeAddress LIKE '%Road%';

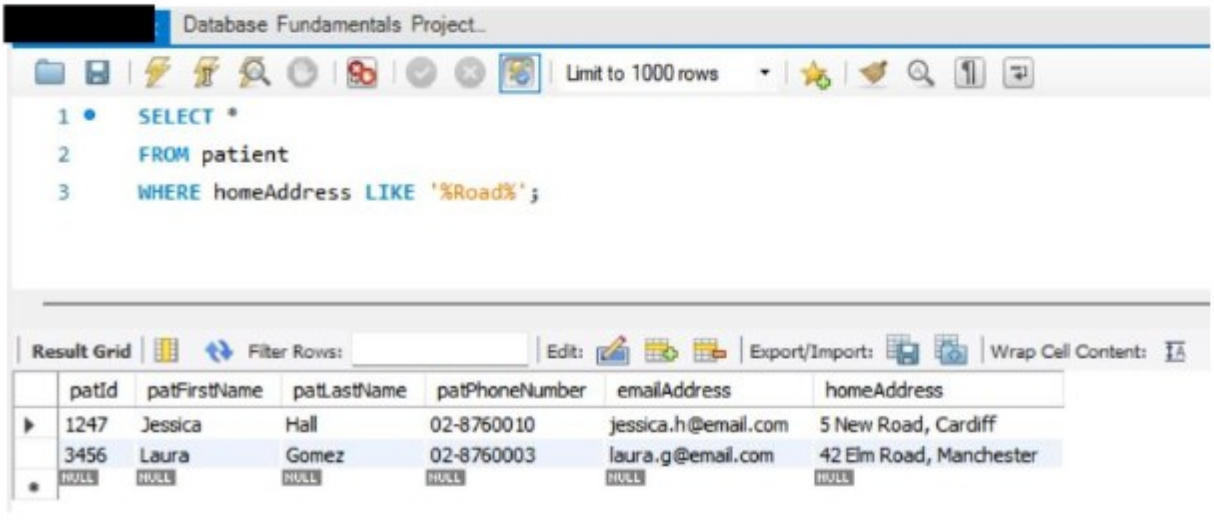
[redacted]




| PatientsWithRoadInAddress |
|---|
| 2 |

## Proof

[redacted]

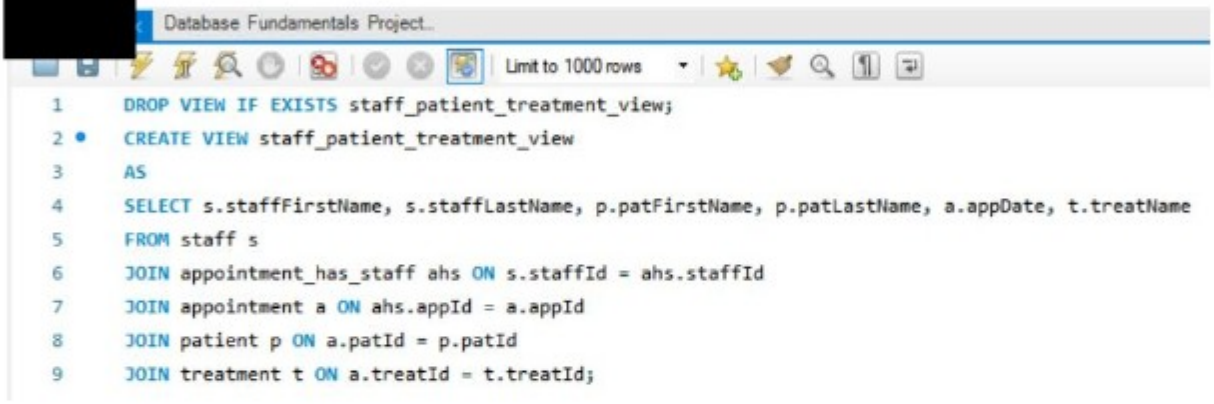



| patId | patFirstName | patLastName | patPhoneNumber | emailAddress | homeAddress |
|---|---|---|---|---|---|
| 1247 | Jessica | Hall | 02-8760010 | jessica.h@email.com | 5 New Road, Cardiff |
| 3456 | Laura | Gomez | 02-8760003 | laura.g@email.com | 42 Elm Road, Manchester |
| NULL | NULL | NULL | NULL | NULL | NULL |

## Create a view that will show the result of a query drawing information from three tables at once.

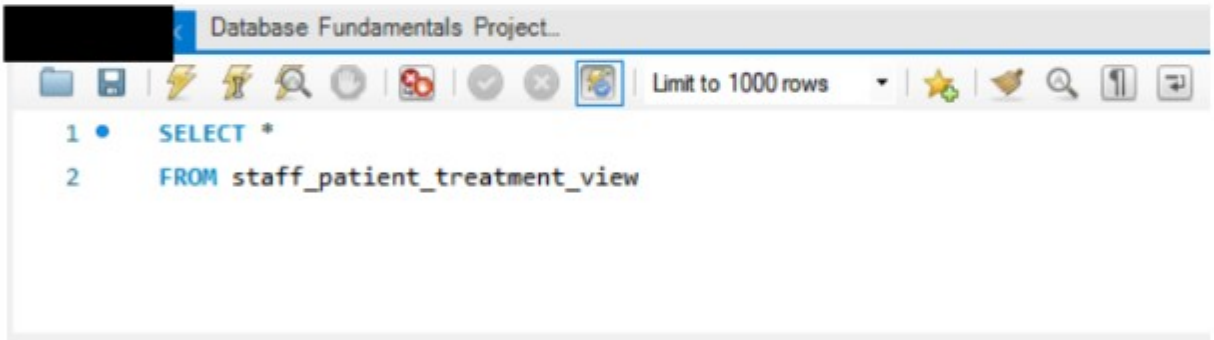
```
DROP VIEW IF EXISTS staff_patient_treatment_view;
CREATE VIEW staff_patient_treatment_view
AS
SELECT s.staffFirstName, s.staffLastName, p.patFirstName, p.patLastName, a.appDate, t.treatName
FROM staff s
JOIN appointment_has_staff ahs ON s.staffId = ahs.staffId
JOIN appointment a ON ahs.appId = a.appId
JOIN patient p ON a.patId = p.patId
JOIN treatment t ON a.treatId = t.treatId;
```

[redacted]




## Proof

[redacted]

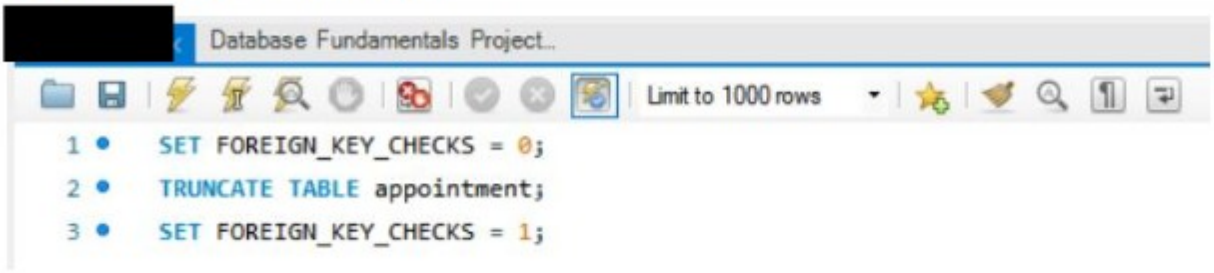



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| staffFirstName | staffLastName | patFirstName | patLastName | appDate | treatName |
|---|---|---|---|---|---|
| Kavin | Smith | Alice | Smith | 2025-10-10 | Heart Surgery |
| Serena | Khan | Robert | Jones | 2025-09-05 | Knee Replacement |
| Jasmine | Lee | Laura | Gomez | 2025-09-15 | Routine Vaccination |
| Marcus | Wright | Michael | Brown | 2025-12-01 | Diagnostic Imaging |
| Olivia | Chen | John | Bob | 2025-12-20 | Spinal Tap Procedure |
| Rohan | Patel | Sarah | Wilson | 2025-12-25 | MRI Scan |
| Amelia | Scott | James | Miller | 2025-08-10 | Biopsy Analysis |
| Ben | Carter | Emily | Clark | 2025-09-01 | Medication Dispensing |
| Chloe | Hayes | Daniel | Walker | 2025-10-05 | Mechanical Ventilation |
| David | Evans | Jessica | Hall | 2025-12-02 | Trauma Stabilization |

## Drop all information contained in the appointments relation.

```
SET FOREIGN_KEY_CHECKS = 0;
TRUNCATE TABLE appointment;
SET FOREIGN_KEY_CHECKS = 1;
```




## Proof

